

COURSE CODE: CP-1259

Python Programming



Diploma in Computer Engineering | Semester 2



Total Hours: 75 | Credits: 4 | Weeks: 15



Revision Date: 27 July 2026

Interactive Lesson with Malayalam Notes • Calculators • Visual Guides • Practice Problems



Source Declaration

This lesson file consolidates the **Python Programming** syllabus (CP-1259) for Diploma in Computer Engineering. The syllabus data is sourced from the official curriculum document. All four modules are covered with detailed explanations, code examples, and bilingual Malayalam-English notes.

Sparse Experiment Details: Some advanced subtopics (e.g., detailed regex flags, all built-in exceptions, advanced metaclasses) are treated as "sparse" — core concepts are fully explained while exhaustive enumeration is deferred to the *References* section. The focus is on *teachable depth* suitable for diploma-level students.



കുറിപ്പ്: ഈ പാഠഭാഗം ഔദ്യോഗിക സിലബസ് അനുസരിച്ച് തയ്യാറാക്കിയതാണ്. എല്ലാ നാല് മൊഡ്യൂളുകളും പൂർണ്ണമായും ഉൾപ്പെടുത്തിയിട്ടുണ്ട്. സാങ്കേതിക പദങ്ങൾ ഇംഗ്ലീഷിൽ തന്നെ സൂക്ഷിച്ചിരിക്കുന്നു.



Course Dashboard

Module	Topic	Hours	Weightage	Status
1	Python Basics — Data Types, Operators, Control Flow, Strings	18	25%	✓ Complete
2	Data Structures — Lists, Tuples, Dictionaries, Sets	18	25%	✓ Complete
3	Functions & Modules — Functions, Recursion, File I/O, Modules	20	25%	✓ Complete
4	Advanced Topics — OOP, Exception Handling, Regex, Debugging	19	25%	✓ Complete
Total			100%	✓



How to Use This Lesson

1. **Start with the Roadmap:** Understand the learning path.
2. **Read each Module sequentially:** Modules build upon each other.
3. **Try the code examples:** Copy-paste into your Python interpreter or IDE.
4. **Use the calculators:** Interactive tools to test concepts.
5. **Check Malayalam notes (.ml-note boxes):** For conceptual clarity in your native language.
6. **Attempt Practice Questions:** At the end of each module and in the dedicated section.
7. **Use Search (🔍):** To quickly find any topic, function, or concept.
8. **Print or Download:** For offline study.



ഉപയോഗിക്കുന്ന വിധം: ഓരോ മൊഡ്യൂളും ക്രമമായി പഠിക്കുക. കോഡ് ഉദാഹരണങ്ങൾ സ്വയം ടൈപ്പ് ചെയ്ത് റൺ ചെയ്യുക. കാൽക്കുലേറ്ററുകൾ ഉപയോഗിച്ച് ആശയങ്ങൾ പരീക്ഷിക്കുക.



Course Outcomes (COs)

Upon completion of this course, the student will be able to:

CO1: Write, debug, and execute Python programs using basic syntax, data types, and operators.

CO2: Implement control flow using conditionals and loops, and manipulate strings effectively.

CO3: Utilize Python's built-in data structures — lists, tuples, dictionaries, and sets — to solve problems.

CO4: Design modular programs using functions, recursion, modules, and handle file I/O operations.

CO5: Apply object-oriented programming principles including classes, inheritance, and polymorphism.

CO6: Handle errors using exception handling, use regular expressions, and apply debugging techniques.



Syllabus Coverage Map

Module	Topics Covered	Depth
1	Introduction, Identifiers, Data Types, Operators, <code>if-elif-else</code> , <code>for</code> / <code>while</code> loops, <code>break</code> / <code>continue</code> / <code>pass</code> , String methods, String formatting	Deep
2	Lists (creation, methods, slicing, comprehensions), Tuples (immutability, packing/unpacking), Dictionaries (keys, values, methods, dict comprehensions), Sets (operations, methods)	Deep
3	Function definition, arguments (default, keyword, *args, **kwargs), scope, recursion, lambda, map/filter/reduce, modules (<code>import</code> , <code>math</code> , <code>random</code> , <code>datetime</code>), File I/O (read, write, append, <code>with</code>)	Deep
4	OOP (classes, objects, <code>__init__</code> , inheritance, polymorphism, encapsulation), Exception handling (<code>try-except-finally</code>), Regular expressions (<code>re</code> module), Debugging (<code>pdb</code> , logging, assertions)	Deep (regex & pdb sparse)



Learning Roadmap

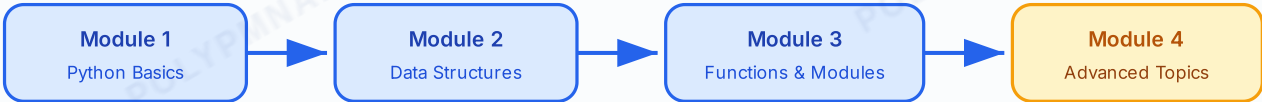


Figure: Sequential learning path through the 4 modules of Python Programming.



Module 1: Python Basics

Hours: 18 | **Topics:** Introduction, Data Types, Operators, Control Flow, Strings

1.1 Introduction to Python

Python is a high-level, interpreted, dynamically-typed programming language created by **Guido van Rossum** in 1991. It emphasises code readability with its use of significant whitespace (indentation).



പൈത്തൺ പരിചയം: വളരെ ലളിതമായി പഠിക്കാവുന്ന ഒരു പ്രോഗ്രാമിംഗ് ഭാഷയാണ് പൈത്തൺ. ഇൻഡന്റേഷൻ വളരെ പ്രധാനമാണ് — ബ്രാക്കറ്റുകൾക്ക് പകരം സ്പേസുകൾ ഉപയോഗിച്ചാണ് കോഡ് ബ്ലോക്കുകൾ നിർവചിക്കുന്നത്.

```
# Your first Python program
print("Hello, World!") # Output: Hello, World!

# Python is dynamically typed
x = 42 # integer
x = "hello" # now a string — perfectly valid!
```

1.2 Identifiers & Data Types

Identifiers are names given to variables, functions, classes, etc. Rules: must start with a letter or underscore, can contain letters/digits/underscores, case-sensitive, and cannot be a reserved keyword.

Core Data Types:

Type	Example	Mutable?	Description
<code>int</code>	<code>42</code>	✗	Integer (unlimited precision)
<code>float</code>	<code>3.14</code>	✗	Floating-point number
<code>str</code>	<code>"hello"</code>	✗	String (sequence of characters)
<code>bool</code>	<code>True</code>	✗	Boolean (<code>True</code> / <code>False</code>)
<code>list</code>	<code>[1,2,3]</code>	✓	Ordered, mutable collection
<code>tuple</code>	<code>(1,2,3)</code>	✗	Ordered, immutable collection
<code>dict</code>	<code>{"a":1}</code>	✓	Key-value pairs
<code>set</code>	<code>{1,2,3}</code>	✓	Unordered, unique elements

1.3 Operators

Python supports arithmetic (`+` `-` `*` `/` `//` `%` `**`), comparison (`=` `≠` `<` `>` `≤` `≥`), logical (`and` `or` `not`), assignment (`=` `+=` `-=` `*=` `/=`), identity (`is` `is not`), and membership (`in` `not in`) operators.

```
a, b = 10, 3
print(a + b) # 13
print(a ** b) # 1000 (10³)
print(a / b) # 3.333...
print(a // b) # 3 (Floor division)
print(a % b) # 1 (modulus)

# Logical operators
print(a > 5 and b < 5) # True
print(not False) # True

# Membership
fruits = ["apple", "banana", "mango"]
print("banana" in fruits) # True
print("grape" not in fruits) # True
```

1.4 Control Flow — Conditionals


```
score = 85
if score ≥ 90:
    grade = "A"
elif score ≥ 80:
    grade = "B"
elif score ≥ 70:
    grade = "C"
else:
    grade = "F"
print(f'Grade: {grade}') # Grade: B
```

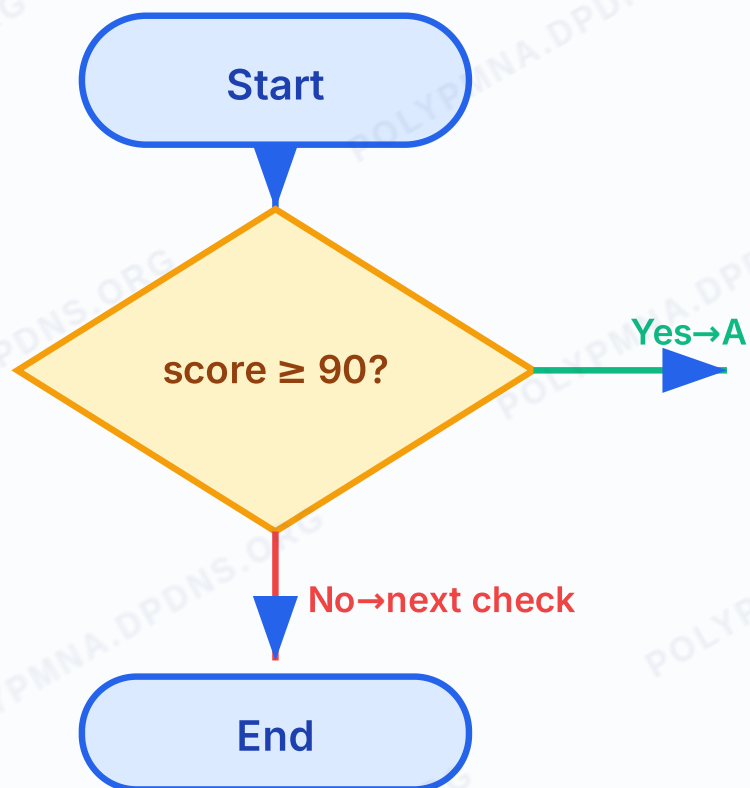


Figure: Flowchart of if-elif-else conditional logic for grade assignment.

1.5 Loops — `for` and `while`

```
# for loop with range
for i in range(1, 6):
    print(i, end=" ") # Output: 1 2 3 4 5

# Iterating over a list
colors = ["red", "green", "blue"]
for color in colors:
    print(color.upper()) # RED GREEN BLUE

# while loop
count = 0
while count < 5:
    print(count, end=" ")
    count += 1
# Output: 0 1 2 3 4

# break and continue
for num in range(10):
    if num == 3:
        continue # skip 3
    if num == 7:
        break # stop at 7
    print(num, end=" ")
# Output: 0 1 2 4 5 6
```

⚠ Common Mistake: Forgetting to update the loop variable in a **while** loop leads to an **infinite loop**. Always ensure the condition eventually becomes **False**.

1.6 Strings

Strings are immutable sequences of Unicode characters. Python provides rich string methods.

```
text = " Hello, Python World!"
print(text.strip()) # "Hello, Python World!"
print(text.lower()) # " hello, python world! "
print(text.upper()) # " HELLO, PYTHON WORLD! "
print(text.replace("Python", "Java")) # " Hello, Java World! "
print(text.split(",")) # [' Hello', ' Python World! ']
print("-".join(["a", "b", "c"])) # "a-b-c"

# String formatting
name, age = "Aarav", 20
print(f"My name is {name} and I am {age} years old.")
# Output: My name is Aarav and I am 20 years old.

# Slicing
s = "Python"
print(s[0:3]) # "Pyt"
print(s[-1:1]) # "nohtyP" (reversed)
```



Module 2: Data Structures

Hours: 18 | **Topics:** Lists, Tuples, Dictionaries, Sets

2.1 Lists

Lists are **ordered**, **mutable** collections. They are the workhorse data structure in Python.

```
# Creating lists
empty_list = []
numbers = [1, 2, 3, 4, 5]
mixed = [1, "hello", 3.14, True]

# List methods
numbers.append(6)      # [1,2,3,4,5,6]
numbers.insert(0, 0)    # [0,1,2,3,4,5,6]
numbers.remove(3)       # [0,1,2,4,5,6]
popped = numbers.pop()  # popped=6, list=[0,1,2,4,5]
numbers.sort(reverse=True) # [5,4,2,1,0]
print(len(numbers))     # 5

# Slicing (creates a new list)
print(numbers[1:4])     # [2,3,4]

# List comprehension
squares = [x**2 for x in range(1, 6)]
print(squares)          # [1, 4, 9, 16, 25]

# Nested lists (2D)
matrix = [[1,2,3], [4,5,6], [7,8,9]]
print(matrix[1][2])     # 6
```

Memory diagram: numbers = [5, 4, 2, 1, 0]

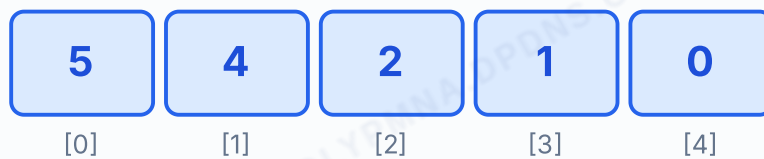


Figure: Memory layout of a list — each element stored at a contiguous index.

2.2 Tuples

Tuples are **ordered**, **immutable** sequences. Once created, they cannot be modified.

```
# Creating tuples
point = (3, 4)
single_item = (42,) # comma is essential!
empty = ()

# Tuple unpacking
x, y = point
print(x, y) # 3 4

# Why tuples? Faster than lists, hashable (can be dict keys), safe from modification
coordinates = {(0,0): "origin", (1,0): "right"}
print(coordinates[(1,0)]) # "right"

# Converting between list and tuple
my_list = [1, 2, 3]
my_tuple = tuple(my_list)
back_to_list = list(my_tuple)
```

⚠ Common Mistake: `t = (42)` is NOT a tuple — it's just the integer `42` in parentheses. Use `t = (42,)` with a trailing comma for a single-element tuple.

2.3 Dictionaries

Dictionaries store **key-value pairs**. Keys must be immutable (strings, numbers, tuples). Values can be anything.

```
# Creating dictionaries
student = {"name": "Aarav", "age": 20, "grade": "A"}
empty_dict = {}
using_constructor = dict(name="Priya", age=21)

# Access and modify
print(student["name"]) # "Aarav"
print(student.get("city", "N/A")) # "N/A" (safe access)
student["age"] = 21
student["city"] = "Kochi"

# Dictionary methods
print(student.keys()) # dict_keys(['name', 'age', 'grade', 'city'])
print(student.values()) # dict_values(['Aarav', 21, 'A', 'Kochi'])
print(student.items()) # dict_items([('name', 'Aarav'), ...])

# Dictionary comprehension
squares_dict = {x: x**2 for x in range(1, 6)}
print(squares_dict) # {1:1, 2:4, 3:9, 4:16, 5:25}

# Iterating
for key, value in student.items():
    print(f"{key} → {value}")
```

Hash Table: student = {"name": "Aarav", "age": 21}

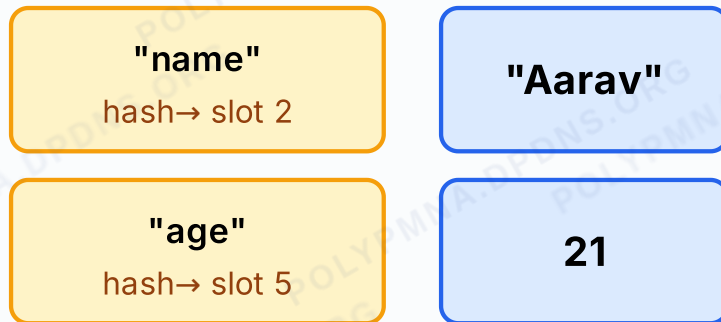


Figure: Dictionary uses a hash table — keys are hashed to determine storage slots for $O(1)$ average lookup.

2.4 Sets

Sets are **unordered**, **mutable** collections of **unique**, **hashable** elements. Great for membership testing and eliminating duplicates.

```
# Creating sets
fruits = {"apple", "banana", "mango"}
empty_set = set() # NOT {} — that creates a dict!
nums = set([1, 2, 2, 3, 3, 4])
print(nums) # {1, 2, 3, 4} — duplicates removed!

# Set operations
a = {1, 2, 3, 4}
b = {3, 4, 5, 6}
print(a | b) # Union: {1,2,3,4,5,6}
print(a & b) # Intersection: {3,4}
print(a - b) # Difference: {1,2}
print(a ^ b) # Symmetric difference: {1,2,5,6}

# Set methods
fruits.add("grape")
fruits.remove("banana")
print("mango" in fruits) # True (fast O(1) lookup)
```

Module 3: Functions & Modules

Hours: 20 | **Topics:** Functions, Recursion, Lambda, Modules, File I/O

3.1 Defining Functions

```
def greet(name, greeting="Hello"):
    """Return a greeting string.""" # docstring
    return f"{greeting}, {name}!"

print(greet("Aarav")) # Hello, Aarav!
print(greet("Priya", "Namaste")) # Namaste, Priya!

# *args and **kwargs
def sum_all(*args):
    """Accept any number of positional arguments"""
    return sum(args)

print(sum_all(1, 2, 3, 4, 5)) # 15

def print_info(**kwargs):
    """Accept any number of keyword arguments."""
    for key, value in kwargs.items():
        print(f"{key}: {value}")

print_info(name="Aarav", age=20, city="Kochi")
```

3.2 Scope & Lifetime

Python follows the **LEGB** rule: **L**ocal → **E**nclosing → **G**lobal → **B**uilt-in.

```
x = "global" # global scope

def outer():
    x = "enclosing" # enclosing scope
    def inner():
        x = "local" # local scope
        print(f"inner: {x}")
    inner()
    print(f"outer: {x}")

outer()
print(f"global: {x}")
# Output:
# inner: local
# outer: enclosing
# global: global
```

⚠ Common Mistake: Using a mutable default argument like `def f(lst=[]):` — the same list object is shared across all calls!
Use `def f(lst=None): if lst is None: lst = []` instead.

3.3 Recursion


```
def factorial(n):
    """Return n! using recursion."""
    if n <= 1:
        return 1 # base case
    return n * factorial(n - 1) # recursive case

print(factorial(5)) # 120

# Tracing factorial(3):
# factorial(3) → 3 * factorial(2)
# factorial(2) → 2 * factorial(1)
# factorial(1) → 1 (base case)
# Unwinding: 1 → 2*1=2 → 3*2=6
```



Output Trace:

```
factorial(3) = 3 * factorial(2)
             = 3 * (2 * factorial(1))
             = 3 * (2 * 1)
             = 3 * 2 = 6
```

3.4 Lambda & Higher-Order Functions

```
# Lambda: anonymous single-expression function
square = lambda x: x ** 2
print(square(5)) # 25

# map: apply function to every element
nums = [1, 2, 3, 4, 5]
doubled = list(map(lambda x: x * 2, nums))
print(doubled) # [2, 4, 6, 8, 10]

# filter: keep elements that satisfy condition
evens = list(filter(lambda x: x % 2 == 0, nums))
print(evens) # [2, 4]

# reduce: cumulative operation (needs functools)
from functools import reduce
total = reduce(lambda acc, x: acc + x, nums, 0)
print(total) # 15
```

3.5 Modules

```
# Importing standard library modules
import math
print(math.sqrt(16)) # 4.0
print(math.pi) # 3.141592653589793

from random import randint, choice
print(randint(1, 100)) # random integer 1-100
print(choice(["red", "green", "blue"])) # random pick

import datetime
now = datetime.datetime.now()
print(now.strftime("%Y-%m-%d %H:%M:%S"))
# Output: 2026-07-27 14:30:00 (example)
```

3.6 File I/O

```
# Writing to a file
with open("sample.txt", "w", encoding="utf-8") as f:
    f.write("Hello, Python!\n")
    f.write("Line 2\n")

# Reading from a file
with open("sample.txt", "r", encoding="utf-8") as f:
    content = f.read()
    print(content)

# Reading line by line
with open("sample.txt", "r", encoding="utf-8") as f:
    for line in f:
        print(line.strip())

# Appending
with open("sample.txt", "a", encoding="utf-8") as f:
    f.write("Appended line\n")

# The 'with' statement auto-closes the file — no need for f.close()
```

📌 **ഫയൽ I/O:** `with` സ്റ്റേറ്റ്‌മെന്റ് ഉപയോഗിക്കുന്നതാണ് ഏറ്റവും സുരക്ഷിതം. ഇത് ഫയൽ ഓട്ടോമാറ്റിക്കായി ക്ലോസ് ചെയ്യും — `f.close()` വിളിക്കാൻ മറന്നാലും പ്രശ്നമില്ല.



Module 4: Advanced Topics

Hours: 19 | **Topics:** OOP, Exception Handling, Regex, Debugging

4.1 Object-Oriented Programming (OOP)

```

class Animal:
    """Base class for all animals."""
    species_count = 0 # class variable

    def __init__(self, name, species):
        self.name = name # instance variable
        self.species = species
        Animal.species_count += 1

    def speak(self):
        return f"{self.name} makes a sound."

    @classmethod
    def get_count(cls):
        return cls.species_count

    @staticmethod
    def kingdom():
        return "Animalia"

# Inheritance
class Dog(Animal):
    def __init__(self, name, breed):
        super().__init__(name, "dog") # call parent __init__
        self.breed = breed

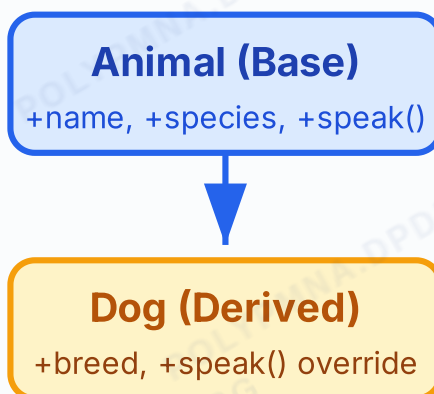
    def speak(self): # override
        return f"{self.name} barks!"

# Polymorphism
animals = [Animal("Generic", "unknown"), Dog("Bruno", "Labrador")]
for a in animals:
    print(a.speak())

# Output:
# Generic makes a sound.
# Bruno barks!

print(Animal.get_count()) # 2
print(Animal.kingdom()) # Animalia

```



Inheritance ►

Figure: Inheritance hierarchy — Dog inherits from Animal and overrides speak().

4.2 Exception Handling

```
def safe_divide(a, b):
    try:
        result = a / b
    except ZeroDivisionError:
        print("Error: Cannot divide by zero!")
        return None
    except TypeError:
        print("Error: Invalid types for division.")
        return None
    else:
        print("Division successful!")
        return result
    finally:
        print("Cleanup: safe_divide finished.")

print(safe_divide(10, 2))  # 5.0
print(safe_divide(10, 0))  # None
# Output:
# Division successful!
# Cleanup: safe_divide finished.
# 5.0
# Error: Cannot divide by zero!
# Cleanup: safe_divide finished.
# None
```

4.3 Regular Expressions (Sparse)

```
import re

text = "Contact: aarav@email.com or priya123@college.edu"

# Find all email patterns
pattern = r'[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}'
emails = re.findall(pattern, text)
print(emails)  # ['aarav@email.com', 'priya123@college.edu']

# Search and replace
cleaned = re.sub(r'\d+', 'XXX', "Phone: 9876543210")
print(cleaned)  # Phone: XXX

# Match validation
phone_pattern = r'^\d{10}$'
print(re.match(phone_pattern, "9876543210"))  # Match object
print(re.match(phone_pattern, "12345"))  # None
```

Note: Advanced regex flags and lookahead/lookbehind assertions are considered sparse topics — refer to Python [re](#) module documentation for exhaustive coverage.

4.4 Debugging Techniques

```
# Using assertions
def calculate_average(numbers):
    assert len(numbers) > 0, "List cannot be empty"
    return sum(numbers) / len(numbers)

# Using logging
import logging
logging.basicConfig(level=logging.DEBUG,
                    format='%(levelname)s: %(message)s')

def process_data(data):
    logging.debug(f"Processing {data}")
    result = [x * 2 for x in data]
    logging.info(f"Result: {result}")
    return result

process_data([1, 2, 3])
# Output:
# DEBUG:Processing [1, 2, 3]
# INFO:Result: [2, 4, 6]

# pdb (Python Debugger) - sparse intro
# import pdb; pdb.set_trace() # set breakpoint
```

🔴 **ഡീബഗ്ഗിംഗ്:** `print()` ഉപയോഗിച്ച് ഡീബഗ് ചെയ്യുന്നതിനു പകരം `logging` മൊഡ്യൂൾ ഉപയോഗിക്കുന്നതാണ് നല്ല ശീലം. ഇത് ലെവൽ അനുസരിച്ച് മെസേജുകൾ ഓൺ/ഓഫ് ചെയ്യാൻ സഹായിക്കുന്നു.



Formula & Syntax Bank

Concept	Syntax / Formula	Example
List comprehension	<code>[expr for item in iterable if condition]</code>	<code>[x**2 for x in range(5) if x%2==0]</code>
Dict comprehension	<code>{k: v for k,v in iterable}</code>	<code>{x:x**2 for x in range(3)}</code>
Lambda	<code>lambda args: expression</code>	<code>lambda a,b: a+b</code>
Ternary operator	<code>val_if_true if cond else val_if_false</code>	<code>"Even" if n%2==0 else "Odd"</code>
f-string	<code>f"text {variable}"</code>	<code>f"Age: {age}"</code>
File open	<code>with open(path, mode) as f:</code>	<code>with open("f.txt","r") as f:</code>
Exception	<code>try: ... except X as e: ... finally: ...</code>	See Module 4



Visual Library

All diagrams in this lesson are embedded as inline SVGs for clarity and accessibility.

- **Figure 1:** Learning Roadmap — sequential module flow (see [Roadmap](#))

- **Figure 2:** if-elif-else Flowchart (see [Module 1.4](#))
- **Figure 3:** List Memory Layout (see [Module 2.1](#))
- **Figure 4:** Dictionary Hash Table (see [Module 2.3](#))
- **Figure 5:** OOP Inheritance Hierarchy (see [Module 4.1](#))



Interactive Activities & Calculators



Grade Calculator

Enter a score (0–100) to see the grade using if-elif-else logic.

Score:

Result will appear here...



Factorial Calculator (Recursive)

Enter a non-negative integer to compute its factorial.

n:

Result will appear here...



BMI Calculator

Enter weight (kg) and height (m) to calculate BMI.

Weight (kg):

Height (m):

Result will appear here...



Assessment Scheme

Component	Weightage	Type
Internal Assessment (IA)	40%	Assignments, quizzes, lab work, attendance
End Semester Exam (ESE)	60%	Written theory + programming problems
Total	100%	Minimum pass: 40% overall

? Practice Questions

Module 1

- 1. Write a program to check if a given number is prime.
- 2. Explain the difference between `break` , `continue` , and `pass` with examples.
- 3. Write a program to reverse a string without using `[::-1]` .

Module 2

- 1. Create a list of 10 numbers and write code to find the second largest element.
- 2. Given two dictionaries, merge them into one. If keys overlap, sum their values.
- 3. Demonstrate set operations (union, intersection, difference) with real-world examples.

Module 3

- 1. Write a recursive function to compute the nth Fibonacci number.
- 2. Explain `*args` and `**kwargs` with a practical example.
- 3. Write a program to count word frequency in a text file.

Module 4

- 1. Design a `BankAccount` class with deposit, withdraw, and balance-check methods.
- 2. Write a function that uses exception handling to safely parse an integer from user input.
- 3. Use regex to validate Indian mobile numbers and email addresses.



Model Question Paper

Duration: 3 Hours | Max Marks: 100

Part	Questions	Marks
A	10 short-answer (all compulsory)	10 × 2 = 20
B	5 out of 7 medium-length	5 × 8 = 40
C	2 out of 4 long programming problems	2 × 20 = 40

Sample Questions:

- **Part A:** Define tuple unpacking. What is the output of `print(2**3**2)` ?
- **Part B:** Explain dictionary methods with examples. Write a program using list comprehension to filter even numbers.
- **Part C:** Design a complete class hierarchy for a Library Management System. Write a Python program to read a CSV file and generate a summary report.



Revision Notes

Module 1 Quick Recap

- Python is dynamically typed and uses indentation for blocks.
- Core types: int, float, str, bool, list, tuple, dict, set.
- `if-elif-else` for branching; `for` and `while` for loops.
- Strings are immutable; use methods like `.strip()`, `.split()`, `.join()`.

Module 2 Quick Recap

- Lists: mutable, ordered. Use `.append()`, slicing, comprehensions.
- Tuples: immutable, ordered. Good for fixed data and dict keys.
- Dicts: key-value pairs. Use `.get()` for safe access.
- Sets: unique elements. Use `|`, `&`, `-`, `^` for set operations.

Module 3 Quick Recap

- Functions: `def`, default args, `*args`, `**kwargs`.
- Scope: LEGB rule (Local→Enclosing→Global→Built-in).
- Recursion needs a base case to terminate.
- File I/O: use `with open(...) as f:` for auto-closing.

Module 4 Quick Recap

- OOP: Classes, `__init__`, inheritance, polymorphism, encapsulation.
- Exceptions: `try-except-else-finally`.
- Regex: `re.findall()`, `re.sub()`, `re.match()`.
- Debugging: Use `logging` instead of `print()`.



Glossary

Argument: A value passed to a function when it is called.

Class: A blueprint for creating objects, defining attributes and methods.

Dictionary: An unordered, mutable collection of key-value pairs with O(1) average lookup.

Exception: An error that occurs during program execution, which can be caught and handled.

Immutable: An object whose state cannot be modified after creation (e.g., str, tuple, int).

Lambda: An anonymous, single-expression function defined with the `lambda` keyword.

List Comprehension: A concise way to create lists using a single line of code: `[expr for item in iterable]`.

Module: A file containing Python definitions and statements that can be imported.

Mutable: An object whose state can be modified after creation (e.g., list, dict, set).

Recursion: A technique where a function calls itself to solve smaller instances of the same problem.

References

- Python Official Documentation: docs.python.org/3/
- Python Standard Library: docs.python.org/3/library/
- PEP 8 — Style Guide: peps.python.org/pep-0008/
- Real Python Tutorials: realpython.com
- W3Schools Python: w3schools.com/python/
- GeeksforGeeks Python: geeksforgeeks.org